

11. Emulator on your AAPS smartphone

V 3.2

Please note that with autoISF you are in an early-dev. environment, where the user interface is **not optimized for safety** of users who stray away from intended ways to use. Good safety features exist, but these are only as good as the development-oriented user understands and implements them. This is not a medical product, refer to disclaimer in [section 0](#)



11.1 Installing the emulator on your smartphone

- 11.1.1 Download QPython 3L
- 11.1.2 .py files in phone internal memory
- 11.1.3 .config file
- 11.1.4 .vdf files
- 11.1.5 Customization of output table

11.2 Checking loop decisions on the smartphone

- 11.2.1 Principal purpose
- 11.2.2 Generating **results table**
- 11.2.3 Analyzing results
- 11.2.4 Console running in background

11.3 Options available on i-Phone (Trio or iAPS)

11.4 Real-time checking a „what-if“ question using **speech synthesis**

[Available related case studies:](#)

Case study 11.1: none available yet

Skip what is in green writing:

= Drafted fragments or not implemented ideas.
Please contribute, or wait for update with the missing info

The emulator **on the PC** was presented in [section 10](#) as

- a very good tool for making your **initial** tuning for a meal spectrum (= when weighing different effects over the entire course of time after each meal, and for a variety of your meals).

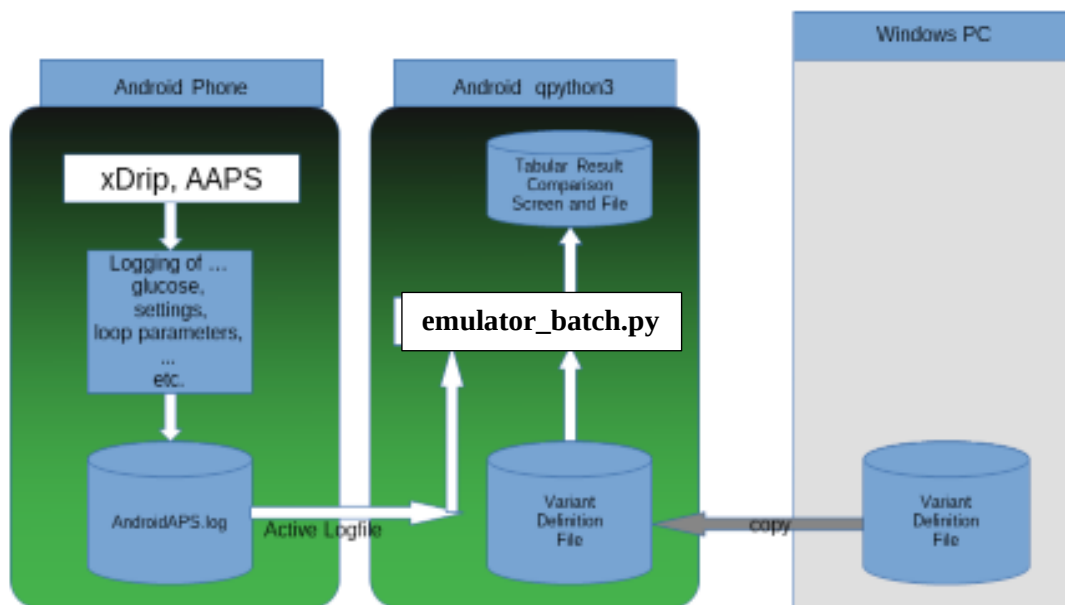
A very useful *additional* tool, is the emulator running **on the AAPS loop smartphone**

- for a quick look how autoISF triggers SMBs after starting a meal ([section 11.2](#)).
- and especially for real-time checking „what-if“ you implemented a specific change idea (developed on your PC, or after analyzing many AUTO ISF tabs) ([section 11.4](#)).

Regarding **i-Phone** options, see [section 11.3](#)

The emulator for the AAPS phone is described in <https://github.com/ga-zelle/APS-what-if>

Join <https://discord.gg/n3tD5eXExC> for seeking (and giving) help with the Emulator set-up or use, and to share experience.



Sketch of Running the Emulator on AAPS Phone

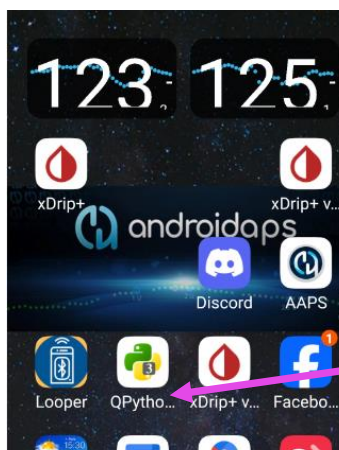
Github/ga-zelle /
APS what-if

11.1 Installing the emulator on your Android smartphone

Note that iOS based autoISF variants cannot use the Emulator on an i-Phone. However, some tabular outputs of ISF-factors are available (see [section 11.3](#))

11.1.1 Installing QPython 3L

On your smartphone, go to Google Playstore and download the QPython 3L app. Put the app icon next to your other looping related app icons on the main screen of your smartphone.



With many QPython 3L versions, phone and Android OS versions etc around, you might run into problems and may need to consult detailed installation instructions from https://github.com/ga-zelle/APS-what-if/blob/A3.2.0.4_ai3.0.1/Documentation%20in%20English/Installation%20Guide.pdf, or seek advice via <https://discord.gg/n3tD5eXExC>

By long pressing on the QPython 3L app icon, go into “app info” and make the settings like for all your other looping related apps, so they do not get killed all the time by energy savings routines.

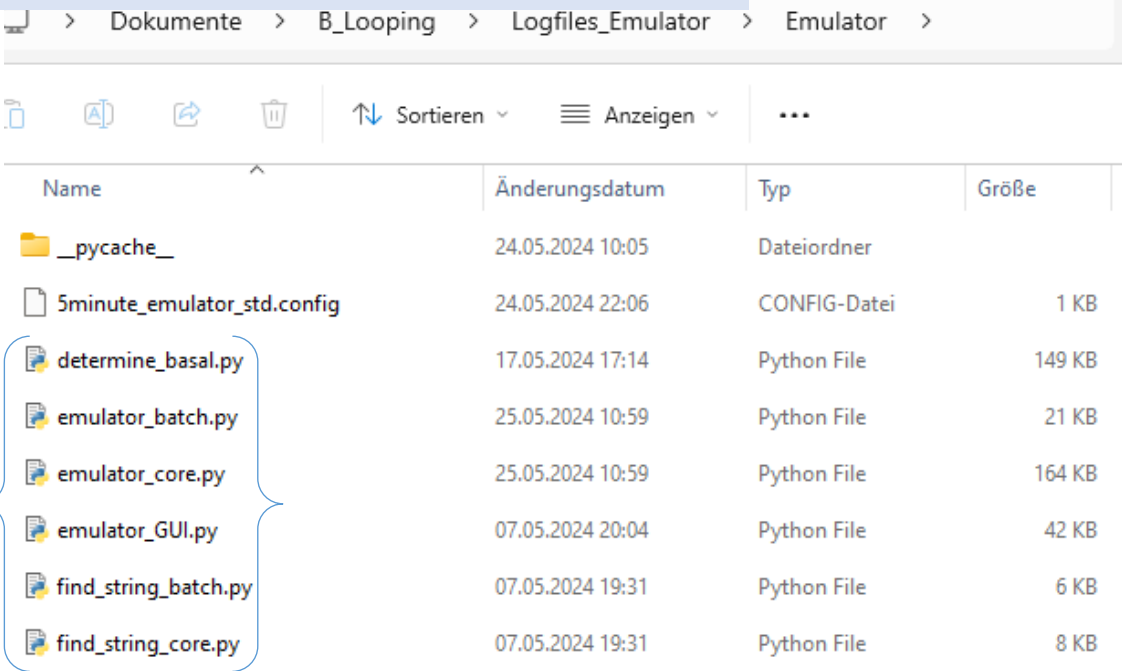
11.1.2 Copy .py files from your PC into your phone's QPython/skript3 folder:

1). Connect your phone to your PC for USB data transfer

67 2).Locate your .py files on your PC (in Emulator file).

68

PC: - your path to the Emulator files may differ -



Name	Änderungsdatum	Typ	Größe
__pycache__	24.05.2024 10:05	Dateiordner	
5minute_emulator_std.config	24.05.2024 22:06	CONFIG-Datei	1 KB
determine_basal.py	17.05.2024 17:14	Python File	149 KB
emulator_batch.py	25.05.2024 10:59	Python File	21 KB
emulator_core.py	25.05.2024 10:59	Python File	164 KB
emulator_GUI.py	07.05.2024 20:04	Python File	42 KB
find_string_batch.py	07.05.2024 19:31	Python File	6 KB
find_string_core.py	07.05.2024 19:31	Python File	8 KB

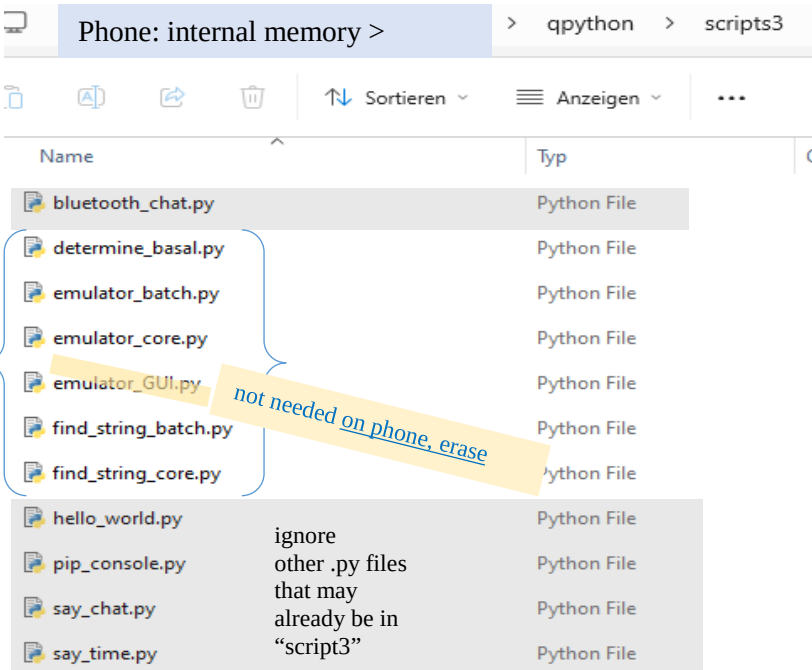
69

70

71 3).Copy all Python related files except emulator:GUI.py from your PC over the internal memory /

72 QPython / Scripts3 of your phone:

73



Name	Typ
bluetooth_chat.py	Python File
determine_basal.py	Python File
emulator_batch.py	Python File
emulator_core.py	Python File
emulator_GUI.py	Python File
find_string_batch.py	Python File
find_string_core.py	Python File
hello_world.py	Python File
pip_console.py	Python File
say_chat.py	Python File
say_time.py	Python File

ignore other .py files that may already be in "script3"

not needed on phone, erase

74

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76

77

78

79

80 11.1.3 Put configuration file and noChange.vdf into the phone's AAPS logfiles folder

81

82 1). With your phone connected to your PC for data transfer, retrieve **5minute_emulator.std.config**

83 (or 1minute_emulator.std.config if you use Libre3/1minute) on your PC, (picture, left)

84

85 2). This config file contains **your "STANDARD"** programming as to which hours of day there can be speech

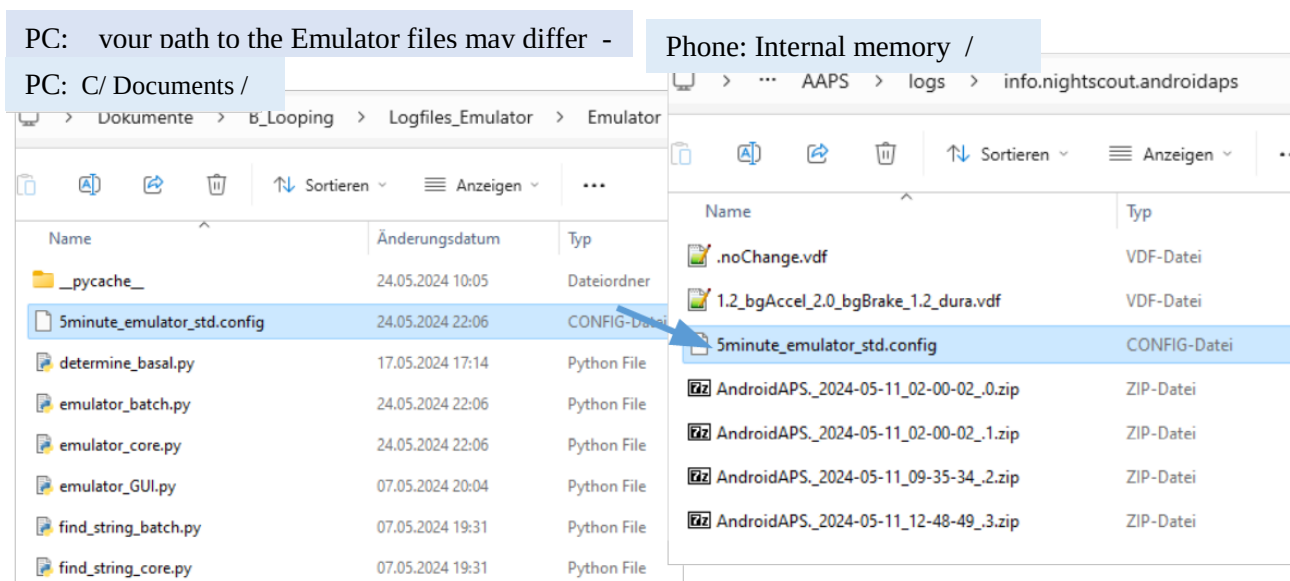
86 messages. Default 07-23 h ("your" time, not "Z"scale). How to change: see in [section 11.4.3](#)

87

88 3). Put a **copy** into the phone into the logfiles (not the QPython!) folder (picture, right):

89 • Internal memory/AAPS/logs/info.nightscout.androidaps

90



91

92 4). You have the option to produce more than just your „...std.config“ file.

93 For instance you could additionally define and load one, that remains silent at carb-related messages, and

94 gives you only insulin (SMB size) related “what-if” suggestions via speech synthesis ; name it for instance

95 „5m_noCarbsAnnounced.config“ .

96 How to switch between the .config files in a run, see [section 11.2.2](#) step 6)

97

98 11.1.4 Put noChange.vdf into the phone's AAPS logfiles folder

99

100 Retrieve the **noChange.vdf**. on PC in the parallel Studies file

101 • C:\....\Documents\ Looping\ Logfiles_Emulator\ Emulator_Studies

102

103 ...and put it on the phone, also into the folder (picture, right):

104 • Internal memory/AAPS/logs/info.nightscout.androidaps

105

106 Later, in [section 11.4.1](#), you will add also (yourChange).vdf files into the same folder. One is
107 already included, line under noChange.vdf, in picture above, right.

108

109 11.1.5 Customization of output table

110

111 The table should contain the most relevant information that can be displayed reasonably on most
112 smartphone screens

113 Consult [https://github.com/ga-zelle/APS-what-](https://github.com/ga-zelle/APS-what-if/blob/A3.2.0.4_ai3.0.1/Documentation%20in%20English/How-to-run-the-emulator-on-the-phone.pdf)

114 [if/blob/A3.2.0.4_ai3.0.1/Documentation%20in%20English/How-to-run-the-emulator-on-the-](https://github.com/ga-zelle/APS-what-if/blob/A3.2.0.4_ai3.0.1/Documentation%20in%20English/How-to-run-the-emulator-on-the-phone.pdf)

115 [phone.pdf](https://github.com/ga-zelle/APS-what-if/blob/A3.2.0.4_ai3.0.1/Documentation%20in%20English/How-to-run-the-emulator-on-the-phone.pdf) in case you see a need to customize .config files and output tables for you.

116

117

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121 11.2 Inspecting loop decisions on your smartphone

122

123 11.2.1 Principal purpose

124

125 The result table from the emulator on your smartphone allows you much easier insight
126 than the AUTO ISF tab can offer into current and recent determinants of given SMB sizes
127 (e.g. which of the 4 autoISF categories contributed).

128 So, if you would have benefitted from a bigger (or smaller) SMB at times where, say,
129 bgAccel_SF was the dominant factor, you would increase (or decrease) the associated
130 weight. Before actually making such changes, look in other lines of the table to estimate
131 how this would influence decisions in other time points.

132

133 Testing your hypothesis on the PC ([section 10.3](#)) would allow multi-day multi-(kind-of-)-meal
134 judgement on feasibility of your tuning idea.

135

136 Before firmly deciding on a settings change, it is advised to run on your phone a what-if
137 emulation ([section 11.3](#)) using the less aggressive settings for your active loop run, and the
138 more aggressive defined in (yourChange).vdf.

139

140

141 11.2.2 Generating a results table (for last 75 minutes' loop decisions)

142

143 Note: 1 minute **Libre3 users** will get data only on a significantly shorter time span.

144 15 minutes is really too short to analyze what is going on, and Libre3 users probably should
145 mostly use the “what-if” part, see [section 11.4](#). That part is principally not impacted (except,
146 cutting one 5 minute change into 5 very tiny changes, often will make it hard to see and
147 interpret “what-if” effects).

148

149 Display setting on your phone should be set for automatic switch between landscape and
150 upright viewing (depending how you hold your phone).

151 Reducing selected font size will *not* help to get more table info on your phone screen, or to
152 avoid broken/double lines. Go to [11.1.4](#) if, in the following, you are not happy with lay-out.

153

154

155 1). On your main phone screen, press the QPython 3L app icon:

156

157 The first black screen then popping up asks to make a **language** selection

158 In case you don't get an alpha-numerical input field (with <- enter button), touch the
159 upcoming black screen

160

161 All black screens have a keypad at the bottom:

162

163

164 2). Press your selected number and <- (for enter).

165

166 3). Then proceed in the same screen

167 to make your capital-”N” entry

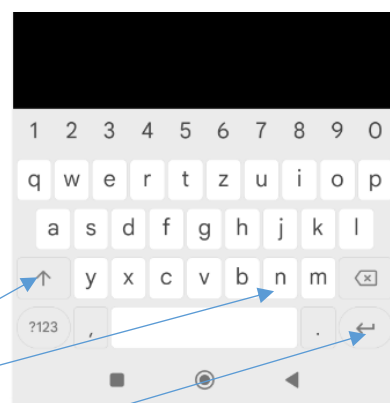
168 and finally “ <- “ for Next (see illustration)

169

170

171

172 In the following the instructions from the developer how to navigate through the screens:



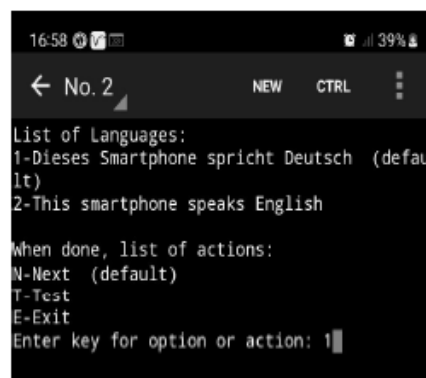
After some Android12 update the previously used GUI dialogues no longer worked and I had to create a keyboard based version similar to the typical telephone dialogues “for option X dial 2”. This dialog system consists of two parts:

- The top part has numerical keys for each option that can be selected. One of those options has **(default)** at the end of the line which indicates that this option is the current selection. If you enter a different number the dialog screen is redrawn and that indicator moved to your new selection. Once your intended selection is OK you focus on the ...
- bottom part, which has letters as keys for the action to be selected. Again, the **(default)** indicator highlights what would be done next if you just press enter without any digit or any letter. Those actions typically are Next, Test and Exit.

The first dialogue is used to select the **language** for the speech synthesis.

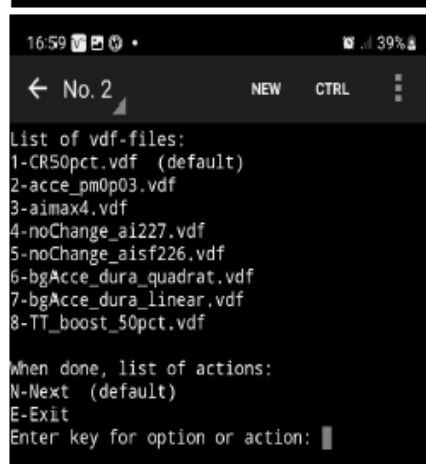
Select „Test“ to listen to a sample speech synthesis.

Select „Next“ to proceed.



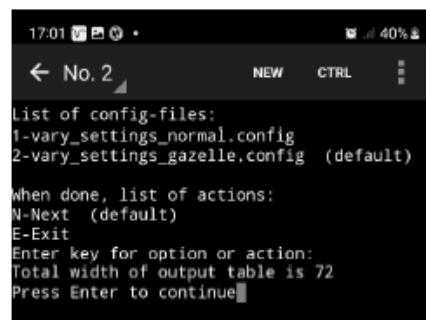
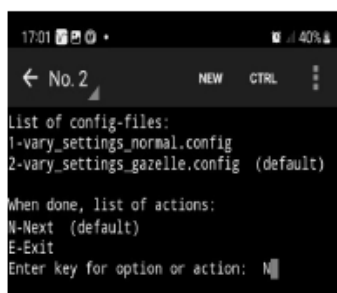
4-5). Repeat steps 2) and 3) with the next screen:

The next dialogue is for selecting your **variant definition file**. All vdf-files found in the logfile folder will be listed.



173 6-7). Repeat steps 2) and 3) with the next screen:

The last dialogue is used to select your favourite **configuration file** with the content discussed in the preceeding section. After you activate the selection you get an informational message of how many columns the selected tabular output will occupy.



8) Before activating the “N(ext)” selection now is a convenient time to rotate the phone to landscape to prepare for the multi column result table display.

174
175 Note: Above in step 6), you can also switch between different ...config files, e.g. to silence less
176 important outputs. See also step 4) in [section 11.1.3](#),



177

178

179 9).Now a table comes up detailing the loop decisions on SMB size for the last (15 * 5 =) 75
 180 minutes.

181 In case you forgot to do it in step 4), turn the phone now 90 degrees for landscape format;
 182 in this case, give it 5 minutes for the format to straighten out (after a new value has arrived)

183 In case you want other, or less, columns, see [section 11.1.4](#).

01:39 84 58%

← N

1

2

3

4

5

6

7

8

9

III

UTC

eff.

acce

bg

pp

dura

-----ISFs-----

insulin Req

---SMB---

time

bg

IOB

iobTH

ISF

ISF

ISF

orig

prof

emul

orig

emul

orig

emul

23:23Z

97

0.93

4.0

-0.28

0.65

1

1

375

150

375.0

0

0

0

0

23:24Z

94

0.9

4.0

-0.44

0.61

1

1

375

150

375.0

0

0

0

0

23:25Z

92

0.89

4.0

-0.21

0.58

1

1

375

150

375.0

0

0

0

0

23:26Z

92

0.85

4.0

0.25

0.58

1

1

375

150

375.0

0

0

0

0

23:27Z

91

0.84

4.0

0.38

0.7

1

1

375

150

375.0

0

0

0

0

23:28Z

89

0.82

4.0

0.4

0.67

1

1

372.8

150

372.8

0

0

0

0

23:29Z

88

0.79

4.0

0.43

0.65

1

1

348.7

150

348.7

0

0

0

0

23:31Z

86

0.77

4.0

0.54

0.61

1

1

275.3

150

275.3

0

0

0

0

23:32Z

85

0.74

4.0

0.63

0.6

1

1

248.2

150

248.2

0

0

0

0

23:33Z

90

0.72

4.0

0.63

0.68

1

1

239.8

150

239.8

0

0

0

0

23:34Z

91

0.69

4.0

3.53

0.71

1

1

59.9

150

59.9

0

0

0

0

23:35Z

87

0.67

4.0

0.78

0.64

1

1

234.1

150

234.1

0

0

0

0

23:37Z

85

0.65

4.0

1.08

0.61

1

1

229.7

150

229.7

0

0

0

0

23:38Z

86

0.63

4.0

0.79

0.62

1

1

241.5

150

241.5

0

0

0

0

23:38Z

88

0.62

4.0

1.2

0.65

1

1

192.1

150

192.1

0

0

0

0

Waiting 60sec for next loop at 01:40; Variant "noChange"

184

185 The line at the bottom of the table says the time (hh:mm) when the next bg result and loop
 186 decision are expected. Also it shows the (yourChange).vdf file investigated

187

188 In sequence of the time(Z!), the table consists default of the columns with info on (1) bg (2)
 189 IOB and eff.iobTH (3-6) the adaptation factors on ISF suggested by the 4 autoISF
 190 categories (7) resulting ISF that was used, profile_ISF, and emul ("what-if" ISF) (8)
 191 insulinRequired calculated by your running loop, and "what-if" result (9) same for resulting
 192 SMB

193

194 The columns marked “**orig**”.in the table of results shows the ISF actually used to
195 determine insulinReq and SMB size in the actual run.

196

197 In the columns marked “**emul**” you find the calculated emulation results that is calculated
198 every 5 minutes

199 • If a (yourChange).vdf was clicked (in step 2), see also bottom right of the table), the emul
200 columns show the result, how the investigated changes *would have* changed SMB sizes (strictly
201 always looking at the one decision, in that line of the table).

202 • If only a noChange.vdf was run, the emul columns contain the same results as the orig.
203 columns.

204

205 **Caution when interpreting the values in the acce, bg, pp and dura ISF columns (3-6)**

206

207 The factors given there are always those for the emul run.

208 • So when using (yourChange).vdf, you see in your phone's table exclusively the ISFs that
209 *would result-if* (yourChange) were made.

210 We get to “what-if” projects later, in [section 11.4](#).

211 • Only in the noChange scenario, the values there would be the *orig ones, corresponding to*
212 *what could be seen also in the AUTO ISF tab* at the times

213 You easy recognize whether you run the noChange: it says so at the bottom of the table.

214

215 orig.ISF is called **sens** in the AUTO ISF tab, a couple of lines below “end autoISF”. It is the
216 ISF that replaces, for that one decision you are looking at, the profile ISF (called
217 profile.sens in the AUTO ISF tab),

218 We get back to this topic at end of [section 11.4.2](#) “Understanding how the ISF is emulated by
219 (yourChange) and how SMB or TBR would differ”

220 “

221

222

223 11.2.3 Analyzing the results

17:05





←

No. 1

NEW

CTRL



1

UTC	eff.	acce	bg	pp	dura	-----ISFs-----			insulin Req		---SMB---			
time	bg	IOB	IobTH	ISF	ISF	ISF	ISF	orig	prof	emul	orig	emul	orig	emu
13:53:42Z	116	2.19	6.0	1.54	1.01	1	1	26.1	40	26.1	0	0	0	0
13:58:42Z	120	2.0	6.0	2.08	1.01	1.12	1.04	19.2	40	19.2	0.16	0.16	0.1	0.1
14:03:57Z	117	1.97	6.0	1.37	1.01	1	1.06	28.5	39	28.5	0	0	0	0
14:08:43Z	113	1.81	6.0	1.18	1.0	1	1.08	33	39	33.0	0	0	0	0
14:13:41Z	115	1.64	6.0	1.11	1.01	1	1.1	35.2	39	35.2	0	0	0	0
14:18:42Z	125	1.47	6.0	1.86	1.02	1.3	1	20.9	39	20.9	2.3	2.3	1.7	1.7
14:23:46Z	131	3.19	6.0	1.44	1.03	1.18	1	27	39	27.0	0.26	0.26	0.1	0.1

224 In above example (table), all given SMBs were driven by bgAccel_ISF, when glucose rose.
 225 The biggest SMB in the time we are looking at (actual local time = Z + 2 hours, so we are
 226 looking at late small rises, like 3 hours after lunch) was 1.7 U = 0.74 SMB delivery rate *
 227 2.3 U insulinRequ.

228 The insulinRequ. Is proportionally amplified by the effective ISF, called sens in your AUTO
 229 ISF tab, or ISF“orig” in this table. The amplification of 39 profile_ISF / 20.9 effective_ISF =
 230 1.86 happens to be the dominating bgAccel_ISF amplification factor

231
 232 Note **not** always just the biggest ISF factor “wins”. Consult flowchart in [LINK](#), and
 233 occasionally read the real-time explanations in your AUTO ISF tab as to which other
 234 factors are contributing to the amplification resulting from profile_ISF to effective_ISF
 235 (“orig”, or sens). For instance, the Activity Monitor, or a set %temp. profile, or TT, could
 236 contribute, or also the question whether glucose already decreases.

237
 238 Depending on your selected “safety” settings, you might occasionally bump into
 239 restrictions. **Tuning up factors that make the system bounce into restrictions is a**
 240 **completely useless, and potentially even dangerous, exercise!**

241
 242 Therefore, **for your initial set-up** of parameters ([section 2](#) and [section 4](#) of the FCL e-
 243 book) it is **advisable to not do this analysis** on the smartphone, but **on the PC**, where
 244 you can inspect the complete info on each loop decision (see [section 10](#))..

245
 246 If for some reason you cannot do this on the PC, you must frequently take screenshots (in
 247 very many decisive 5 minute segments), and analyze **more complete data**, than the table
 248 on our smartphone could offer, **in the AUTO ISF tab**)

249

250 In your initial set-up of your FCL, you probably must “loop” a couple of times back into
251 [section 2](#) to adjust the safety settings made for max. SMB sizes.
252
253 **11.2.4 Console running in background**
254
255 When you are finished analyzing real-time coming up tables, you can either close the
256 emulation program, or (in a box “Run console in background”) you can opt for letting it run
257 in the background.
258
259 This is required if you want the speech synthesis give information at times when a bigger
260 SMB would have been suggested by the (yourChange).vdf emulation. Also you might hear
261 suggestions by when you should eat how many g of carbs (more see in [section 11.4.3](#)).

262

263 **11.3 Options available in Trio or iAPS**

264

265

266 iAPS / Trio offer in their autoISF variants also

267 access to a tabular representation of

268 **autoISF contributions to resulting SMB sizes:**

269

270

271

272

273

274 The emulator will not work in the iOS world.

275

276 Note that on the iPhone, so far, the what-if emulation

277 and speech synthesis (see next [section 11.4](#) for AAPS)

278 are currently **not** possible. .

279

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286 **More description must be provided by iAPS / Trio user please**

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291

autoISF History								Sulje	
30.5.2024 13.00 2 hours ↕									
Time	BG	ISF factors		bg	pp	dura	Insulin		req.
		final	acce				SMB	TBR	
12.16	6,5	0,5	0,1	1	1	1	0	0	0
12.11	7,2	1,07	1	1,01	1	1,07	0	0	0
12.06	7,3	1,07	1,07	1,01	1,07	1,05	0	0	0
12.00	7,1	1,01	1	1	1	1	0	0	0
11.54	7,5	1,18	1	1,02	1	1,18	0	0	0
11.48	7,9	1,22	1	1,04	1	1,22	0	0	0
11.40	7,9	1,18	1	1,04	1	1,18	0	0	0
11.30	7,9	1,12	1	1,04	1	1,12	0	0	0
11.22	8,1	0,5	0,1	1,05	1	1,07	0	0	0
11.17	8,3	0,75	0,71	1,06	1,02	1	0	0	0

11.4 Real-time checking a „what-if“ question using speech synthesis

The emulator on your **Android** smartphone can help clarify "what if..." you implemented a considered change.

11.4.1 Put a (yourChange).vdf into the phone's AAPS logfiles folder

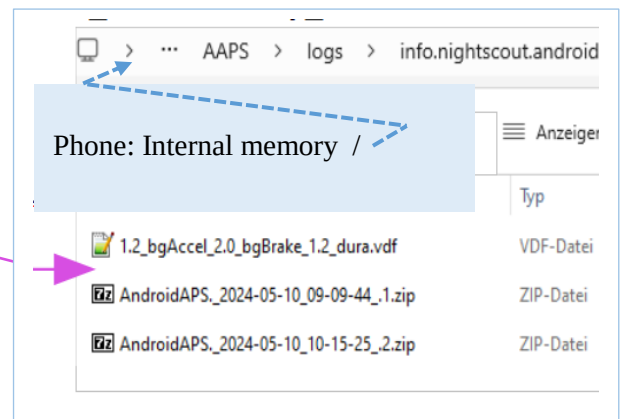
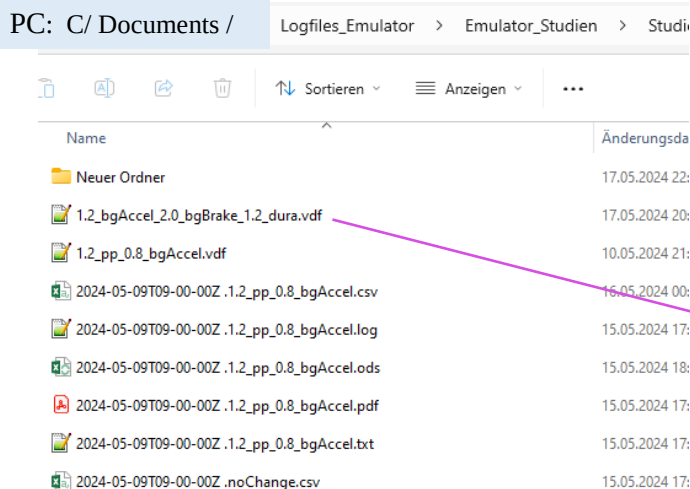
In running the emulator on the phone, you can define in the .vdf file of the emulator, which setting(s) you would like to be differently aggressive than in the active AAPS.

How to write .vdf files was already explained in the section "Emulator on PC". See in end of [section 10.2.1](#). You could also pull a vdf file example from the developer's Github that you could customize further: Access see [section 3.8 /5](#).

Produce or retrieve **(yourChange).vdf**. on PC from one of your studies files.

- C:\....\Documents\ Looping\ Logfiles_Emulator\ Emulator_Studies\ Study_n

PC: your path to the Emulator files may differ -



... and put a **copy** of that .vdf into the **smartphone folder with the AAPS logs** :

- Phone: Internal memory/AAPS/logs/info.nightscout.androidaps

Switch between python scripts running at the same time

In case you have **more than one** (yourChange).vdf to investigate, you can look at the data of your currently running loop (last 15*5 minutes) by just switching between the related vdf files used for emulation.

321 Details see [section 3.8, 9\)](#) – or try to go direct via [https://github.com/ga-zelle/APS-what-](https://github.com/ga-zelle/APS-what-if/blob/A3.2.0.4_ai3.0.1/Documentation%20in%20English/How-to-run-the-emulator-on-the-phone.pdf)
322 [if/blob/A3.2.0.4_ai3.0.1/Documentation%20in%20English/How-to-run-the-emulator-on-the-](https://github.com/ga-zelle/APS-what-if/blob/A3.2.0.4_ai3.0.1/Documentation%20in%20English/How-to-run-the-emulator-on-the-phone.pdf)
323 [phone.pdf](https://github.com/ga-zelle/APS-what-if/blob/A3.2.0.4_ai3.0.1/Documentation%20in%20English/How-to-run-the-emulator-on-the-phone.pdf) and there p.5, under above sub-headline “.Stop the emulator, or switch...”

324

325 [Stop the emulator](#)

326 (see in paper as above)

327

328 [11.4.2 Inspect emulated results](#)

329

330 Now, whenever you run QPython 3L emulation (following the steps as described in [section 11.2.2\)](#)
331 you additionally get the emul. results filled in in the resulting table, showing in which time points
332 your change would lead to adapted insulinReq, and SMB size data.

333

UTC		eff.	acce	bg	pp	dura	-----ISFs-----			insulin Req		---SMB---	
time	bg	IOB	ioBTH	ISF	ISF	ISF	orig	prof	emul	orig	emul	orig	emu
20:41:16Z	112	1.93	6.0	1	1.0	1	1	40.9	41	40.9	0	0	0
20:46:16Z	111	1.77	6.0	1.14	1.0	1	1.03	36	41	36.0	0	0	0

etc.

21:11:17Z	108	1.06	6.0	1.12	1.0	1	1.08	38.3	43	38.3	0	0	0
21:16:18Z	109	0.94	6.0	1.1	1.0	1.03	1.07	39.1	43	39.1	0.1	0.1	0
21:21:20Z	109	0.87	6.36	0.95	1.0	1	1.08	41.9	43	39.5	-0.1	-0.08	0
21:26:20Z	108	0.76	6.36	1	1.0	1	1	43	43	40.6	0	0	0

334

335 etc

336 In the marked 21;21 Z line, dura_ISF was the dominant factor. The (yourChange).vdf would apply a
337 factor of 1.2 and lead to further strengthening the ISF: profile 43 -> orig.(noChange) 41.9 -> emul
338 (yourChange) 39.5

339 In this case, late after a meal, and bg=109 mg/dl, the loop saw in the orig. (noChange) case 0.1 U
340 insulin too much; and as the (yourChange) emul case asks for typically more insulin (all weights in
341 the (yourChange).vdf are > 1), now only 0.08 U are seen as too much (a 20% difference).

342

343 The table on your phone is too reduced to show each emulated ISF component. If you need to see
344 more details on how (yourChange).vdf would make a change in a point of time that you like to
345 analyze deeper: Note down interesting Z times, and later look it up in more detail in the Emulator
346 on the PC:

347

below are "orig."ISFs:

UTC		eff.	acce	bg	pp	dura	-----ISFs-----			insulin Req		---SMB---		
time	bg	IOB	iobTH	ISF	ISF	ISF	ISF	orig	prof	emul	orig	emul	orig	emu
21:21:20Z	109	0.87	6.36	0.95	1.0	1	1.08	41.9	43	39.5	-0.1	-0.08	0	

Understanding ISF orig.(the sensitivity used to determine insulinRequ)

To fully understand how acce, bg, pp and dura_ISF determined the orig.ISF, we do an **emulator** run **on the PC**, and look up the **.txt** result:

@ 21:21 Z / Script Debug -----

Activity monitor disabled inactivity detection: sleeping hours; Autosens ratio: 1;

Basal unchanged: 0.45; ISF unchanged: 43 CR: 9

start autoISF 3.0.1

Loop allows APS power level; SMB enabled due to enableSMB_always

acce_ISF adaptation is 0.95

bg_ISF adaptation is 1

pp_ISF adaptation is 1

dura_ISF adaptation is 1.08 because ISF 43 did not do it for 30 m

strongest autoISF factor 1.08 weakened to 1.03 as bg decelerates already

final ISF factor is 1.03

end autoISF

profile.sens: 43 sens: 41.9

Note: 43 / 1.026 = 41.9

Understanding how the ISF is emulated by (yourChange) and how SMB or TBR would differ

While the result for ISF_emul (and for the SMB size) is given in the table on your phone, the details behind it, and also for finer effects in %TBR, can only be inspected from the .csv table from the (yourChange),vdf **emulator** run on the **PC**:

382 The following table is an extraction of the most relevant data from the (yourChange).csv:

	B	C	E	F	L	Q	R	S	AA	AB	AC	AD	AE	AF	AG	AH	AI	AP	AQ
1			bg	bg		final	dura		acce	bg	pp	delta	dura	final					
2	UTC	AAPS	accel	brake		ISF	min-	dura	ISF	ISF	ISF	ISF	ISF	ISF	ISF	ISF	ISF	TBR	TBR
3	time	time			iob	orig	utes	avg.	emul	emul	emul	emul	emul	emul	orig	prof	emul	orig	emul
29	21:16:18	23:16	109		0,94	1,1	25	108,2	1,12	1	1,03	1	1,08	1,12	39,1	43	38,4	0,65	0,65
30	21:21:20	23:21	109	109	0,87	1,03	30	108,3	0,9	1	1	1	1,1	1,04	41,9	43	41	0	0,46
31	21:26:20	23:26	108		0,76	1	5	108,5	1	1	1	1	1	1	43	43	40,6	0	0
32	C=B+n/24 @ n=+2 h time diff.										If there were bigger orig->emul effects, we would show SMB columns, too								

383

384 It shows that the final (noChange) ISF factor of 1.03 (box Q30) becomes 1.04 (box AF30) in the
 385 (yourChange) emulated case. Given that, so late after dinner (box C30), we sit near the 90 mg/dl
 386 target (box E30), the difference the changes in the emulated case are minor. Still zero SMB, but
 387 the minor changes reflect in elevating the temp. basal rate for fine adjustment (box AP30 -> AQ30).

388

389 Note that the orig. values for the four autoISF components (acce, bg, pp, dura "ISF orig")
 390 are **not** contained in this csv table (only, in Q30, the final amplification factor for final_ISF in
 391 the noChange scenario).

392 However, you can fetch the autoISF details for 21:20 Z in the actual "noChange" run:

- 393 • from the txt result file (as shown above); or
- 394 • you also see them on the phone:

395

orig	orig	orig	orig	ISF
acce	bg	pp	dura	-----
ISF	ISF	ISF	ISF	orig
0.95	1.0	1	1.08	41.9

396

397 ...while the emul_ISFs come from .csv results @ (yourChange).vdf run on the PC:

AA	AB	AC	AE	AI
acce	bg	pp	dura	
ISF	ISF	ISF	ISF	ISF
emul	emul	emul	emul	emul
1,12	1	1,03	1,08	38,4
0,9	1	1	1,1	41,2

398

399

400 Apology: The above example was not well chosen to see relevant effects. The author is struggling to
 401 put this chapter first time together, quickly for the V.3.0.1 launch, and just picked from his phone what
 402 was available at the moment, I might patch it over with a better example in a later update, or I (or
 403 maybe you?) provide an adjunct case study.

404

405

406

407 Principal limitation of any of your emulations

408

409 Note that always the first biggest change regarding insulinRequ and SMB size in emul vs. orig. is
410 the most relevant. This is because:

- 411 ○ Doing that change would change, by the same amount, the iob basis for the next
412 following loop decision
- 413 ○ Doing that change would certainly change the course of the bg curve about half an hour
414 afterwards (to be precise: for the duration of DIA, for the extra insulin), so then it is
415 anybody's guess, for instance how difficult of a job presents itself to the loop to "attack"
416 with duraISF.

417

418 This is one of the reasons why setting up your FCL ([section 4](#)) should be a iterative process,
419 seeking solutions (with mainly bgAccel_ISF) for the first rise (and a range of different meals) first,
420 AND NOT concurrently already "tuning" the dura_ISF.

421 Both, bg peak height, and pattern of insulin activity from the present iob (that is only sluggishly
422 further adjustable driven by dura_ISF and evtl. by bg_ISF) depend on how the first bg rise stage was
423 managed by autoISF (your bgAccel_ISF_weight setting, and others).

424

425 Another conclusion you might draw, is to limit use of the emulator *on the PC* largely to analyzing
426 the (in FCL extremely important) first rise, to seek bgAccel_ and pp_ISF_weights.

427

428 Then to test these, and many other like ideas for changed settings "in real life" emulated *on the*
429 *smartphone*: A synthetic voice will announce to you how (if) each actual decision would differ. See
430 next section!

431

432

433 11.4.3 Real time speech synthesized treatment suggestions

434 Unfortunately this great feature is **not available on i-Phone**. Look in (updates of) [section 11.3](#) for
435 eventual alternatives.

436

437 At time points when the (yourChange) setting would result in smaller or greater difference in SMB
438 insulin delivery (compared to the real "noChange" run), you can get a real-time notification **via**
439 **speech synthesis**, and you can assess the situation in real-time yourself.

440

441 Also if just the noChange.vdf is running (and even in AAPS without autoISF) , the speech synthesis could alert
442 you to "carbs required" messages, for instance.

443

444 If for instance a suggested extra, or bigger, SMB makes sense, **you can add this portion**
445 **manually*and observe**, for this meal, whether this bolus was OK and you should switch in

446 direction of the different setting you were investigating (which would automatically give you that
447 extra in the future).

448 Likewise, you might choose **not** to intervene, but regret it an hour or so later, seeing the further
449 development **without** implementing the supposed improvement.

450

451 **In Full Closed Loop, you don't need any buttons at the bottom of the AAPS main screen. But for such*
452 *test phases it is practical to re-install the insulin button at the bottom of the AAPS main screen*
453 *(Preferences/Overview/Buttons/Insulin -> ON).*

454

455 After a couple of days, you will get a feel for whether you want to incorporate your
456 investigated change (or a gradual step towards it) into your active AAPS settings.

457

458 **Warning:** Your settings must always work for a variety of meals. Do not put too
459 much effort into optimizing one situation! (See [case study 8.2](#)).

460

461 **Activating and silencing emulator suggestions**

462

463 For silencing the suggestions from voice synthesis you have the following options:

464

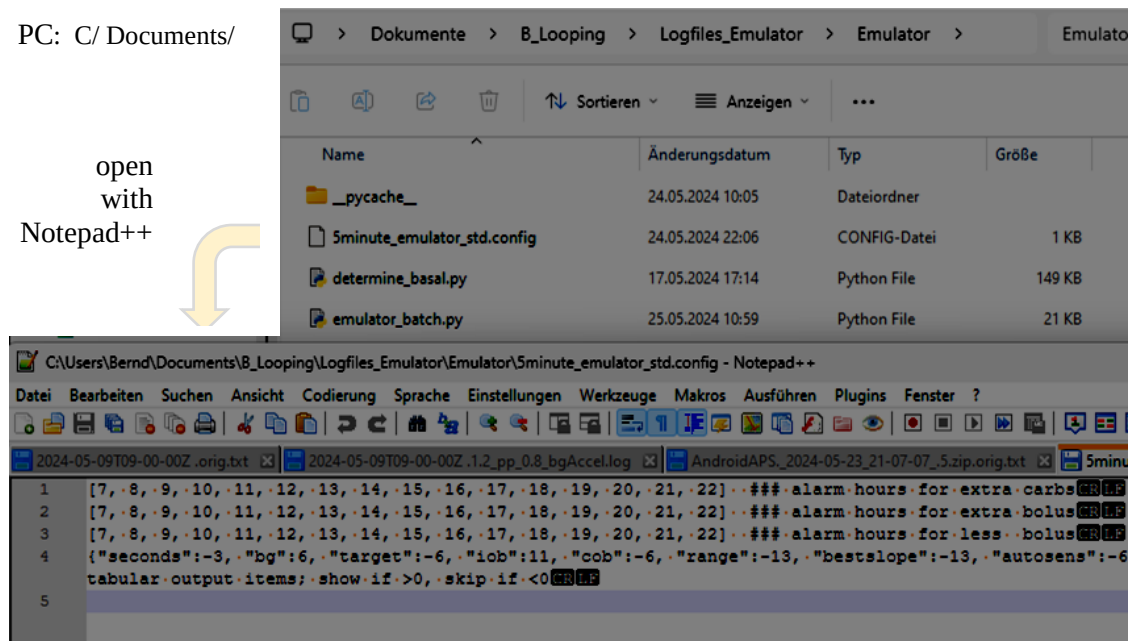
465 (1) Change **principal settings** what shall be announced (e.g. only if bigger SMB size
466 is suggested, or also warnings about carbs eventually needed?), and in **which hours of**
467 **day**, to make any announcements via speech synthesis .

468 These are set in the (1 or) 5minute_emulator_std.config file:

469 Go on **PC** into the config. file (see illustration). Open it with Notepad++ and edit the
470 hours there for when you would want (no) announcements regarding: extra carb
471 need (line 1), extra bolus need (line 2), or less bolus (line 3):

PC: C:/ Documents/

open
with
Notepad++



Save the changes, and copy the file also into your **phone** at Internal memory/AAPS/logs /info.nightscout.androidaps (see [11.1.3](#)) over the 5minute_emulator_std.config

(2) Turn off **phone volume** (silence media + switch on do not disturb)

Of course, this also shuts off many other potential alerts that you might not want to shut off.

(3) **“Kill” (and later resume) the “what-if” emulation.** This could be done by de-selecting the (yourChange).vdf in step . However, this stops (or interrupts, until you re-start) the entire emulation and you will have no tabular data later for the silenced time.

(4) **De-activate Qpython 3L app** (temporarily?): Press on Qpython 3L app icon, force close or remove necessary permission – re-activate (or need to re-start emulation then ??) when you want to hear again .. However, this stops (or interrupts, until you re-start) the entire emulation and you will have no tabular data later for the silenced time

(5) ☐ Probably better than the 2 afore mentioned options would be to **run (for the intended silencing period) exclusively the noChange.vdf**, Then you will not get any speech outputs (because you are NOT investigating a what-if question, in that case). But you get all data (the un-interrupted noChange actual run) and later on the PC still can investigate any “what-if” scenarios

How to change the .vdf reference during a run, see [section 3.8, 9](#) – or go direct to the instructions in the Github repo at: https://github.com/ga-zelle/APS-what-if/blob/A3.2.0.4_ai3.0.1/Documentation%20in%20English/How-to-run-the-emulator-on-the-phone.pdf and there p.5, under above sub-headline “Stop the emulator, or switch...”

498 (6) A variation of option (5) would be to silence all but the (less frequently occurring but
499 most important) one line in your config definition (done according to [section 11.1.5](#))
500 Then go through step 6) in [section 11.2.2](#), you can also switch between different ...config files, e.g.
501 to silence the less important outputs. See also step 4) in [section 11.1.3](#),

502

503 Note that this area (silencing) has not been researched much, and good answers are likely
504 lifestyle and phone specific.

505

506

507 Please share your experiences with the emulator in Discord / Full-Closed-Looping / HOW
508 TO /_emulate-aaps, at: <https://discord.gg/n3tD5eXExC>

509

510

511